

AUTONOMOUS AI-BMS

Safety-Governed, Carbon-Aware AI Building Management System

Problem Statement ID – Question1A

Problem Statement Title – AI-Powered Autonomous Smart Building Optimization

Theme – Smart Buildings / Energy & Sustainability

PS Category – Software

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AUTONOMOUS AI-BMS — PROPOSED SOLUTION

WHAT IT DOES

- Connects an EnergyPlus digital building to a local Qwen LLM supervisory agent.
- Uses zone temperature, occupancy, outdoor weather, HVAC power and grid-carbon context.
- Generates adaptive heating/cooling setpoints every 6 simulated hours.
- Applies approved actions through EnergyPlus HVAC actuators.

HOW IT ADDRESSES THE PROBLEM

- Converts static HVAC scheduling into context-aware autonomous control.
- Maintains occupied thermal comfort while enabling energy/carbon-aware decisions.
- Produces auditable AI decisions and operational metrics.

INNOVATION & UNIQUENESS

- Local LLM: no cloud dependency for core decision-making.
- AI recommendations are never applied directly.
- Semantic Validator checks decision intent and operating context.
- Deterministic Safety Guard enforces physical limits, rate limits and thermostat deadband.
- Real-Time Comfort Guard protects occupants between LLM decisions.
- Dynamic carbon intensity is included in the decision context.
- Fail-safe retains the previous safe action if AI fails.

TECHNICAL APPROACH

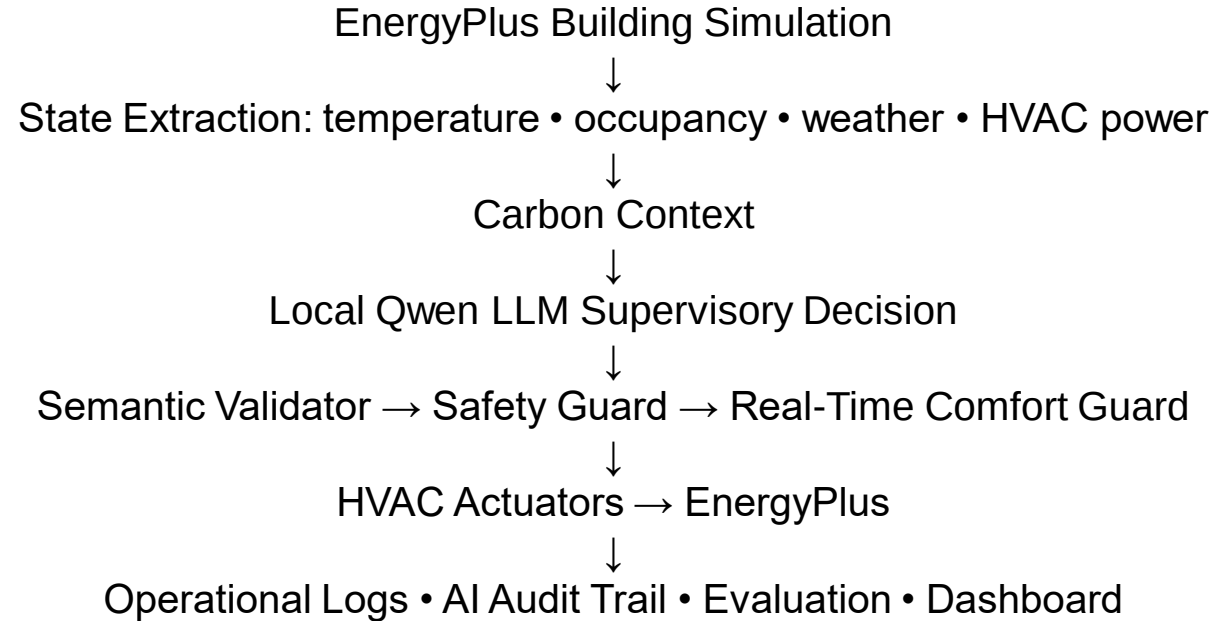
TECHNOLOGIES

- Python
- EnergyPlus 26.1 + Python API
- Local Qwen LLM
- Pandas / CSV analytics
- Streamlit dashboard
- JSON live-state bridge
- Dynamic carbon-intensity model

CONTROL PRIORITY

Safety > Comfort > Energy > Carbon

CLOSED-LOOP METHOD



FEASIBILITY AND VIABILITY

FEASIBILITY

- Working EnergyPlus closed-loop prototype.
- Standard thermostat actuators and measurable building variables.
- Local inference supports privacy and avoids cloud dependency.
- Modular guard layers can transfer to real BMS integrations.

CHALLENGES / RISKS

- LLM latency and malformed outputs.
- Unsafe or inconsistent recommendations.
- Long simulation logs and expensive full-year testing.
- Seasonal energy/comfort trade-offs.
- Winter test identifies conservative heating as an optimization target.

MITIGATION

- 6-hour supervisory AI interval.
- Strict JSON prompt + semantic validation.
- Deterministic safety and real-time comfort guards.
- Fail-safe retains last safe setpoints.
- Hourly operational aggregation + separate AI audit logs.
- Iterative seasonal policy tuning.

ARTIFACTS & VALIDATION

IMPLEMENTED ARTIFACTS

- EnergyPlus autonomous control runner
- Qwen LLM decision engine + engineered prompt
- Semantic decision validator
- Deterministic HVAC Safety Guard
- Real-Time Comfort Guard
- Carbon-aware control model
- Live JSON state / decision feed
- Streamlit monitoring dashboard
- Baseline-vs-AI evaluation pipeline

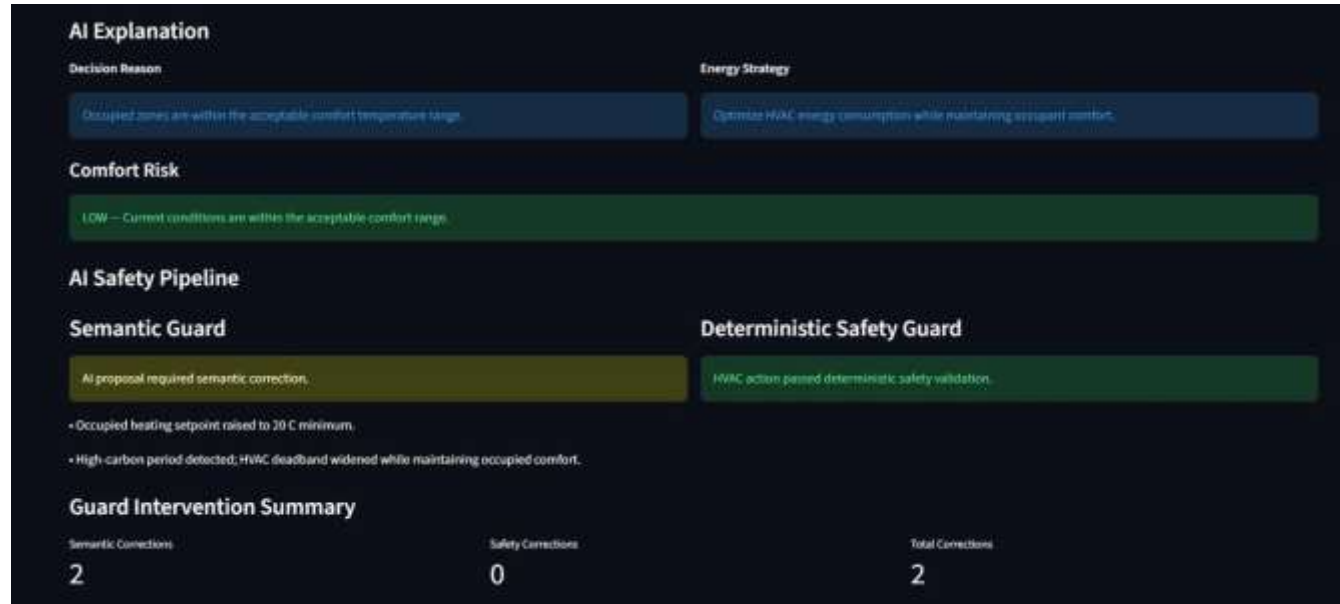
CLEAN 7-DAY WINTER VALIDATION

- Baseline records: 168
- AI records: 168
- Matched records: 161
- Autonomous AI decisions: 28
- Semantic interventions: 15
- Semantic corrections: 19
- Safety interventions: 5
- Safety corrections: 5
- Occupied comfort compliance: 100%

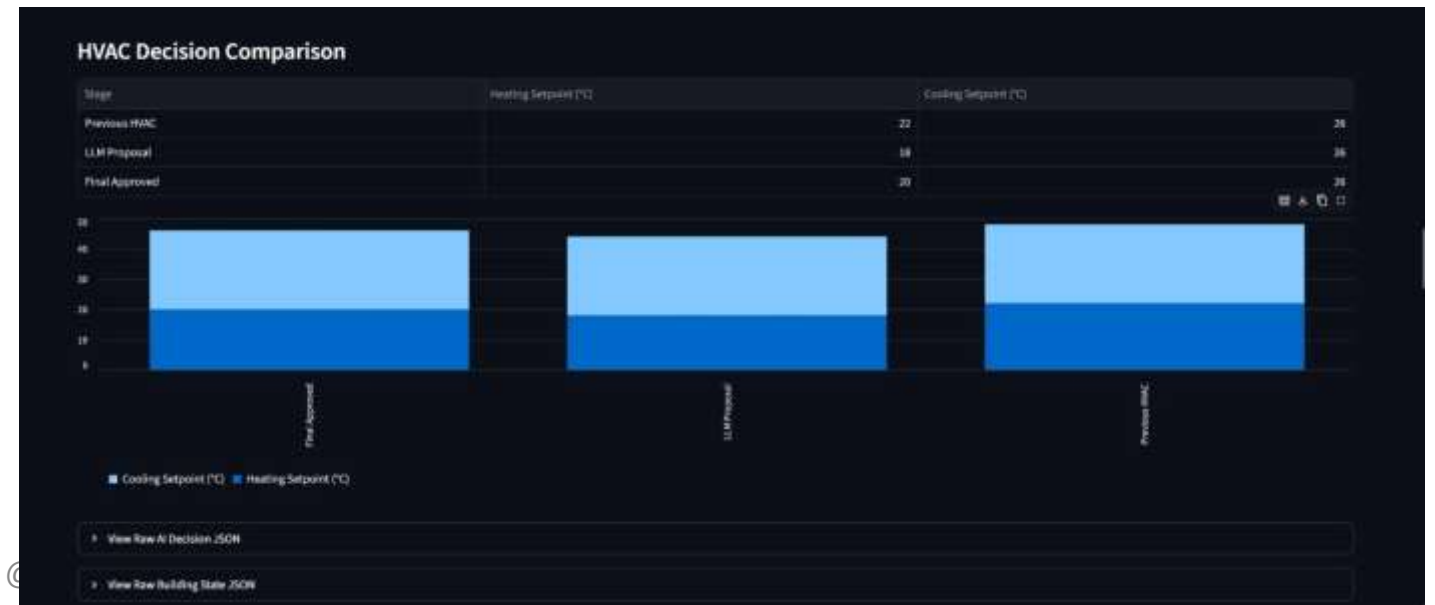
RESULTS



RESULTS(contd...)



All other dashboard results are available in github.



RESEARCH AND REFERENCES

1. U.S. Department of Energy — EnergyPlus Documentation
Simulation engine, Python API, runtime callbacks and data exchange.
2. ASHRAE Standard 55 — Thermal Environmental Conditions for Human Occupancy
Reference for occupant-focused thermal comfort.
3. ASHRAE Guideline 36 — High-Performance Sequences of Operation for HVAC Systems
Reference for robust HVAC control concepts.
4. Qwen — Open-weight Large Language Model family
Local supervisory reasoning agent.
5. Streamlit Documentation
Real-time monitoring and evaluation dashboard.
6. Project experimental evidence
Baseline/AI operational metrics, AI audit logs, semantic/safety interventions and clean 7-day controlled validation.